A BIRATIONAL MORPHISM BETWEEN SCHEMES WITH FINITELY MANY COMPONENTS IS AN ISOMORPHISM OVER THE LOCAL RING OF A GENERIC POINT OF THE TARGET

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 13019-13026 and proof span 13028-13043 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains the author's parenthesis '(small argument omitted)' for the step reducing to an affine open of the source; that phrase is kept verbatim and what it leaves to the reader is the elementary restriction of the claim to an open subscheme, while the main reduction and the final identification are carried out in the counted span.

Allowed standard prerequisites (editorial).

- The definition of a birational morphism: there are dense open subschemes over which the morphism induces an isomorphism, which allows reducing to an affine irreducible case. Used at proof lines 13245-13248. Justifies assuming both schemes affine and irreducible and gives the isomorphism of local rings at the generic points.
- The generic point of an irreducible component corresponds to a minimal prime, and there is a unique minimal prime of the source lying over the minimal prime of the target in the birational situation. Used at proof lines 13248-13252. Produces the primes whose local rings are compared.
- An element of a local ring whose image in the residue field is zero up to the maximal ideal, and the comparison of the localized ring with the local ring at a minimal prime. Used at proof lines 13252-13260. Shows that the localized source ring is a local ring of dimension zero and hence equal to the local ring at the minimal prime.

Lemma 0.1. *Let $f : X \to Y$ be a birational morphism of schemes having finitely many irreducible components. If $y \in Y$ is the generic point of an irreducible component, then the base change $X \times_Y \mathrm{Spec}(\mathcal{O}_{Y,y}) \to \mathrm{Spec}(\mathcal{O}_{Y,y})$ is an isomorphism.*

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We may assume $Y = \mathrm{Spec}(B)$ is affine and irreducible. Then X is irreducible too. If we prove the result for any nonempty affine open $U \subset X$, then the result holds for X (small argument omitted). Hence we may assume X is affine too, say $X = \mathrm{Spec}(A)$. Let $y \in Y$ correspond to the minimal prime $\mathfrak{q} \subset B$. By assumption A has a unique minimal prime $\mathfrak{p}$ lying over $\mathfrak{q}$ and $B_\mathfrak{q} \to A_\mathfrak{p}$ is an isomorphism. It follows that $A_\mathfrak{q} \to \kappa(\mathfrak{p})$ is surjective, hence $\mathfrak{p}A_\mathfrak{q}$ is a maximal ideal. On the other hand $\mathfrak{p}A_\mathfrak{q}$ is the unique minimal prime of $A_\mathfrak{q}$. We conclude that $\mathfrak{p}A_\mathfrak{q}$ is the unique prime of $A_\mathfrak{q}$ and that $A_\mathfrak{q} = A_\mathfrak{p}$. Since $A_\mathfrak{q} = A \otimes_B B_\mathfrak{q}$ the lemma follows. $\square$